

Year: III

Roll No.

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Course Name: B.Sc. Microbiology
Course Code: B-301
Branch(s): Microbiology
COs: CO-1 & CO-2

Pages: 2
Time: 120 Min
Max. Marks: 60

Instructions:

1. Attempt all the sections.
2. Assume missing data suitably, if any.

CO1: On completion of this course, the student will be able to students will be able to understand Immunity and immune system

CO-2: On completion of this course, the student will be able to students will be able to illustrate organs and cells of immune system

KL: Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)
K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Section-A

(2 x 6 = 12 Marks)

Attempt all questions of this section.

Q.No.	Questions	CO	KL
1.	a	1	K1
	b	1	K1
	c	1	K2
	d	2	K1
	e	2	K2
	f	2	K3
Q.No.	Questions	CO	KL
2.	Define Immunity. Describe in detail active and passive immunity.	1	K1
3.	Differentiate between innate and acquired type of immunity.	1	K1
4.	In account of historical development kindly explain the contribution of various scientists in immunology.	1	K1

Section-C

(7 x 2 = 14 Marks)

Attempt any two questions of this section.

5.	Write a note on exogenous and endogenous antigen and their examples.	2	K3
6.	With the help of neat diagram explain a typical structure of antibody in detail.	2	K2
7.	Explain IgG and IgM immunoglobulin on the basis of their typical structure and function.	2	K2

Section-D**Attempt all questions of this section.****(10 x 2 = 20 Marks)**

Q.No.	Questions	CO	KL
8.	Give a comparison among specific and non-specific defence mechanism.	1	K2
	Or		
	“Phagocytic cells are a barrier to infection”, explain. Write a detail note on Antigen and classify the antigen on the basis of their chemical nature, origin and immune response.	1 2	K2 K3
9.	Or		
	Describe the structure of antibody. Also explain its various types and their functions	2	K3